Course - Data Engineering with Python, Azure Databricks & Power BI

What Sets Us Apart:

➢ **Real enterprise-level projects**

➢ **Personalized Work Readiness Assessment (WRA) reports**

➢ **Two rounds of professional mock interview feedback reports**

➢ **Technical skills gap analysis with improvement roadmap**

➢ **Project evaluation reports with industry benchmarking**

➢ **Attendance and participation performance tracking**

➢ **Certification readiness assessment**

➢ **Live recorded lectures**

Day wise Schedule

Day	Topic	Description
Day 1	**Python Basics**	Installation, IDEs, Program Flow, Keywords, Running Scripts
Day 2	**File Operations & Data Structures**	Indexing, Slicing, range(), Dictionaries, Sets, Files
Day 3	**Functions & Advanced Python**	Function Arguments, Parameters, Variable Scope, Decorators, Context Managers

Day 4	**Packages & Error Handling**	Package Installation, Virtual Environments, Exception Handling, Logging
Day 5	**Git & Version Control**	Git fundamentals, GitHub, Branching, Pull Requests, Collaboration
Day 6	**Python Practice & Review**	Hands-on exercises covering Days 1-5, coding challenges
Day 7	**Database Handling**	Python-MySQL Connectivity, CRUD Operations, SQLAlchemy
Day 8	**Database Project**	Building Real-World Application with MySQL
Day 9	**NumPy Fundamentals**	Installation, Arrays, Slicing, Multidimensional Arrays, Broadcasting
Day 10	**NumPy Advanced**	DTypes, Vectorized Operations, Performance Optimization
Day 11	**Pandas Fundamentals**	DataFrames, Series, Indexes, loc[] and iloc[], Reading Data
Day 12	**DataFrame Manipulation**	Accessing Values (at[], iat[]), Filtering, Sorting, Filling & Replacing
Day 13	**DataFrame Operations**	Merging, Joining, Concatenating, Group By Operations
Day 14	**Advanced Pandas**	Pivot Tables, Reshaping, Window Functions, Performance Tips

Day 15	**Time Series & File I/O**	DateTime Operations, Time Series Analysis, Reading/Writing Multiple Formats
Day 16	**Pandas Project Day**	Real-world data cleaning and analysis project
Day 17	**SQL Fundamentals**	Relational Databases, DDL/DML, Query Basics, Joins
Day 18	**SQL Advanced**	Complex Joins, Subqueries, Window Functions, CTEs
Day 19	**SQL Performance**	Query Optimization, Indexes, Execution Plans, Best Practices
Day 20	**SQL Practice Project**	Complex queries, database design, real-world scenarios
Day 21	**Docker Fundamentals**	Containers, Images, Dockerfile, Docker Compose, Basic Orchestration
Day 22	**Apache Spark Introduction**	RDDs, DataFrames, Spark Architecture, Installation & Setup
Day 23	**Spark DataFrames & SQL**	DataFrame API, Spark SQL, Transformations & Actions
Day 24	**PySpark Advanced**	Optimization, Partitioning, Broadcasting, UDFs
Day 25	**Spark Practice Day**	Hands-on large-scale data processing exercises

Day 26	**Apache Kafka Basics**	Streaming Concepts, Producers, Consumers, Topics, Partitions
Day 27	**Kafka & Spark Streaming**	Real-time data processing, Structured Streaming, Integration patterns
Day 28	**Apache Airflow Fundamentals**	DAGs, Operators, Scheduling, Dependencies, Task Management
Day 29	**Airflow Advanced**	Sensors, XComs, Branching, Dynamic DAGs, Best Practices
Day 30	**Airflow Practice Project**	Build end-to-end data pipeline with monitoring
Day 31	**Azure Data Lake Storage (ADLS)**	Cloud Storage Concepts, Data Organization, Access Patterns
Day 32	**Azure Data Factory (ADF)**	Pipeline Building, Copy Activities, Data Flows, Triggers
Day 33	**Databricks Fundamentals**	Workspace, Notebooks, Cluster Management, DBFS, Unity Catalog
Day 34	**Databricks & PySpark**	Large-scale Transformations, Databricks SQL, Performance Optimization
Day 35	**Delta Lake on Databricks**	Delta Lake Architecture, ACID Transactions, Time Travel, Z-Ordering
Day 36	**Databricks Projects**	Data Connection & Transformation Projects, Real-world Scenarios

Day 37	**Databricks Advanced**	Jobs, Workflows, Databricks SQL Analytics, Performance Tuning
Day 38	**Cloud Data Warehouse Basics**	Snowflake/BigQuery Architecture, Best Practices, Cost Optimization
Day 39	**Data Quality & Monitoring**	Great Expectations, Data Validation, Monitoring Pipelines, Alerts
Day 40	**CI/CD for Data Pipelines**	Git workflows, Testing strategies, Deployment automation
Day 41	**Power BI Fundamentals**	Data Connectivity, Visualizations, DAX Basics
Day 42	**Power BI Advanced**	Complex DAX, Performance Optimization, Dashboard Design
Day 43	**End-to-End Project**	Complete Pipeline: Source → ADLS → Databricks → Warehouse → BI
Day 44	**Work Readiness Assessment**	Final Student Interviews with Industry Experts
Day 45	**Work Readiness Assessment**	Final Student Interviews with Industry Experts